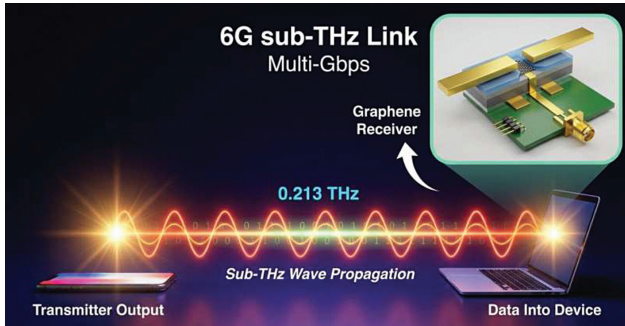


Graphene receivers boost 6G hardware efficiency

Scientists at the Institute of Photonic Sciences (ICFO), Barcelona, have demonstrated graphene-based receivers designed to boost 6G hardware efficiency. Operating at sub-terahertz frequencies, the receivers deliver high

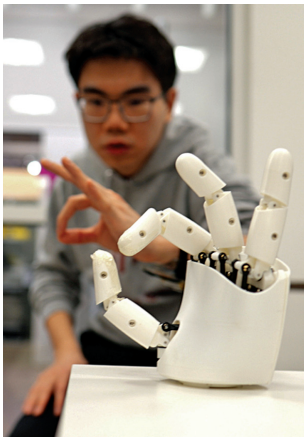


Scheme showing the general idea of the study (Credit: ICFO)

sensitivity while consuming significantly less power. The breakthrough addresses key challenges in next-generation wireless systems, enabling faster data transmission and more compact designs, marking a major step toward practical and scalable 6G communication infrastructure.

Ultrasound wristband enhances human-computer interaction

A wearable ultrasound wristband can track hand movements in real time and translate them into digital or robotic



Graduate student Dian Li working with a robotic hand (Photo Credit: Melanie Gonick)

actions without cameras or gloves. The system captures internal wrist activity and uses artificial intelligence to accurately predict finger and palm positions. Developed at the Massachusetts Institute of Technology, the device images muscles, tendons, and ligaments inside the wrist, enabling precise gesture recognition and control. The technology could enhance human-computer interaction, robotics, and virtual environments by enabling intuitive, contact-

less control. It builds on efforts to overcome limitations of camera-based and sensor-driven hand-tracking systems.

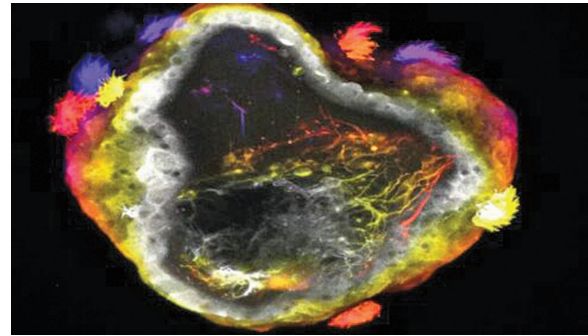
AI stack runs on local system

A fully indigenous AI stack has been successfully deployed by Turiyam AI on a local computing system, demonstrating that advanced AI workloads can run without reliance

on cloud infrastructure. The system integrates the model, inference engine, and compute into a single domestic environment, enabling end-to-end execution on local hardware. Developed through work conducted at the Centre for Development of Advanced Computing (C-DAC) in Pune, the platform combines an Indian-built inference engine with Rudra servers and a large Hindi-language model that supports 37 dialects. The achievement could strengthen data sovereignty, reduce latency, and lower cloud dependency for AI applications. It reflects growing momentum toward self-reliant, edge-based AI ecosystems built entirely on domestic infrastructure.

Living robots built from frog cells

Scientists at the Wyss Institute for Biologically Inspired Engineering at Harvard University have developed 'neurobots', living robots built from frog cells with self-organising nervous systems. Unlike earlier biobots, these machines integrate



Neurobots have a surface of specialised cells and a nervous system controlling motion (Credit: <https://www.msn.com>)

neurons that form active networks, enabling coordinated movement and adaptive behaviour. The breakthrough allows internal control of biological robots, opening pathways for applications in regenerative medicine, environmental cleanup, and advanced biohybrid robotics systems.

Non-flammable battery electrolyte enables safer batteries

A new non-flammable battery electrolyte has been developed, aiming to significantly improve the safety and reliability of lithium-based energy storage systems. The innovation reduces fire risks associated with conventional liquid electrolytes, which are prone to ignition and thermal runaway. Engineered by a research team at University of Tennessee at Knoxville in China, the electrolyte leverages nanoscale self-assembly to balance stability and ionic performance. This design enhances thermal resilience while maintaining efficiency. The breakthrough could impact electric vehicles and portable electronics by enabling safer batteries. It builds on ongoing efforts to replace flammable components in next-generation energy storage technologies.